East Swindon Development Group

WATER VOLE AND OTTER SUMMARY REPORT

November 2010

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Rev	Issue Status	Prepared / Date	Approved/Date
-	Draft 1	DH/18/11/10	PH / 01.02.10

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1.0 INTRODUCTION

1.1. This report has been prepared by FPCR Environment and Design Ltd on behalf of East Swindon Development Group. It follows on from previous FPCR Ecological Reports conducted through the duration of this project, where various ecological surveys have been conducted. This report details the results of an update otter and water vole survey that was carried out on the River Cole, Dorcan Stream and the South Marston Brook, which cross the site.

1.2. Initial surveys for otters and water voles were undertaken in large areas of the site during 2006, where evidence of both species were found, with otter spraints seen around Acorn bridge located just off the A420. Water vole evidence was seen in the north eastern regions of the River Cole, but evidence decreased the further west upstream on the South Marston Brook. During the surveys conducted during September 2010, water vole evidence was seen along the River Cole which runs westerly through the site, however field signs were sparse and very isolated.

1.3. During the survey of the site no otter evidence was seen. The lack of field evidence for both species could be explained by the fluctuating water levels owing to a generally wet summer.

2.0 METHODOLOGY

Background

Water Vole and Otter

2.1. Water voles are now fully protected under section 9 of the Wildlife and Countryside Act 1981 (as amended). Otters are fully protected through their inclusion in Schedule 2 of The Conservation (Natural Habitats, &c.) Regulations 1994 (as amended) and under the Wildlife & Countryside Act 1981 (as amended). They are afforded protection under Section 9 parts 4(a) and 4(b). Water courses were assessed for their potential as suitable habitat for water voles and otters including the degree of bank side and emergent vegetation, bank shape and angle and size and flow of the water course. Water courses were also examined for physical signs of water voles and otters, including:

- Prints;
- Water vole latrines or otter spraints;
- Holes visible in the banks typical of the species;
- Grazed ‘lawns’ along aquatic margins;
- Feeding stations; and
- Possible otter lying up sites such as sheltered cavities in river side willows.

3.0 RESULTS (REFER TO FIGURE 1)

South Marston Brook

3.1. This brook runs through the northern areas of the site and flows west to east where it joins to the River Cole. The majority of the banks were over shaded by hedgerows and trees which bound a number of arable fields; bank side vegetation was also low in abundance. Water levels ranged from 30-60cm in depth and were fast flowing causing a lot of erosion on the soft clay substrate that lines the brook. Some sections of the brook had a stone bed.

3.2. The banks were steep at approximately 70 -90 degrees, which was made up of a mixture of clay and soil, the over hanging root systems helped to support the banks and provided grassy overhangs that would be ideal territory marking locations for water voles. The number of over hanging roots and trunks would also be good locations for otter spraints to be located, however nothing was seen (see photo 1).

3.3. The eastern sections of brook recorded no signs of either otters or water vole, however the fresh abrasions of the banks would suggest that water levels had recently been high, thus this might have washed away any evidence. However the lack of vegetation suggest this brook less suitable for water voles, although the banks do have the potential to be used for burrowing.

3.4. To the west of this ditch was a small tributary that headed south towards the A420 (target note 1 see figure, photo 2), this only had a couple of centimetres of water but supported a wide variety of marginal species such as common comfrey, greater willowherb and meadow sweet, there were also sections of reedmace and common reed. Water levels decreased closer to the A420. During surveys here no evidence of any water vole or otters were seen.

3.5. Further north eastward from target note 1, the habitat continued to improve and water levels were a little more consistent, however no direct evidence of either species were seen. There were prints within the muddy substrate corresponding to previous positive sightings, which showed water vole characteristics, although the weathering of prints made confirmation of species uncertain. This ditch continued north easterly out of the site boundary.

Photo 1: Typical habitat within the South marston Stream

Photo 2: Small ditch running from south to north from the A420

River Cole

3.6. This is a reasonably wide water course, however water levels were shallow and the river walked with care. Water flow was slow over the mixture of sand, clay and occasional rocky bed. Banks were approximately 1 – 2m in height and were formed of a clay that could provide a good burrowing conditions for water voles. In the northern region, north of Acorn Bridge and the A420, no evidence of otters or water voles were seen. Suggesting a decline in activity or lower population density.

3.7. As the River headed southward under the A420, sections became wider with varying water depths and flow rates with corresponding variation in bank profile, some sections had marginal ledges where flow rates were more sluggish, while other areas supported vertical banks either side with faster flowing water (see photo 3). Some marginal vegetation was seen around the bridge near the A420, however no feeding signs were seen.

Photo 3: River Cole

3.8. Overshading by trees and scrub, which lined the majority of the northern bank increased towards the west and marginal vegetation was lacking as a result. A number of prints were seen where marginal shelves had formed. These could not be confirmed as water vole prints, although a few metres west a small groups of animal dropping were found. These were cylindrical in shape and a similar size droppings normally associated with water voles, however these were uncharacteristically a tar black colour, further droppings were less uniformed and were identified as rat *Rattus norvegicus* droppings. Further footprints were seen in scattered location along the eastern section of the river.

3.9. Confimration of water vole occupation was note at the confluence of the River Cole and Liden Brook where a recently established water vole latrine was seen, this had approximately fifteen individual droppings. Small patches of marginal vegetation had established in slow flowing parts of the river, this included watercress, yellow flag iris and common reed provided suitable, if localised, foraging.

3.10. A further water vole latrine was also noted toe the west, this only consisted of ten droppings with no other associated evidence.

3.11. During the entire survey of the river Cole and Marston Brook no evidence of any otters were seen. This species can have an extensive range and it may well be possible that only sporadic use of this area occurs or that evidence may have been washed away during the changing water levels. It was considered likely that otters are still present within the catchment, although use of the site only occasional.

4.0 DISCUSSION

4.1. Water voles Locally, records of this species are relatively widespread. Nevertheless, due to the declines suffered nationally this species is included as a priority species in the UK and LBAP's and is now afforded full protection under the Wildlife and Countryside Act 1981 (as amended), which makes it an offence to:

- Intentionally kill, injure or take water voles
- Possess or control live or dead water voles or derivatives
- Intentionally or recklessly damage, destroy or obstruct access to any structure or place used for shelter or protection
- Intentionally or recklessly disturb water voles whilst occupying a structure or place used for that purpose
- Sell water voles or offer or expose for sale or transport for sale
- Publish or cause to be published any advertisement which conveys the buying or selling of water voles.

4.2. The above Act provides a defence against the above where the action is "the incidental result of an otherwise lawful operation and could not have reasonably been avoided." This therefore requires that attention is paid to the presence of water voles and, as far as is reasonable, appropriate action is taken to safeguard the animals and the places they use for shelter and protection. Whilst it may be possible to rely on this defence it is highly desirable to ensure that water voles are removed from harm where it is possible to do so.

4.3. Licenses are available from Natural England to allow activities that would otherwise be unavailable, although these cannot be granted for development alone they can be granted for conservation reasons, where, for example, a developer considers that the best outcome would be to trap and translocate water voles from working areas. Where this is the case it will be necessary to demonstrate that:

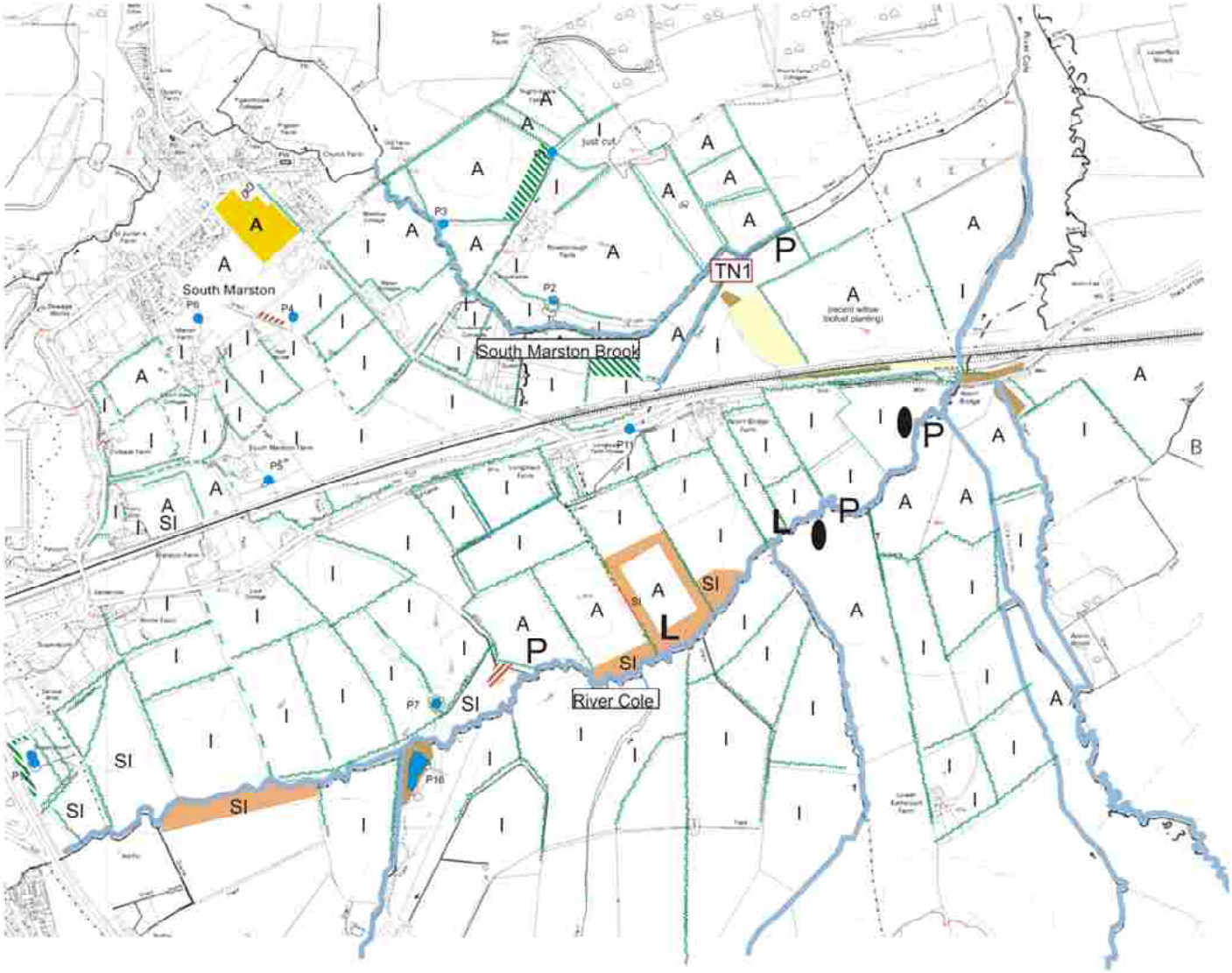
- The activity proposed is lawful i.e. appropriate planning permission or other lawful authority.
- The development and the likely impacts on water voles could not reasonably have been avoided and that all reasonable efforts had been made to carry out the lawful development in a way that minimised the risk of committing an offence.
- The translocation of water voles would provide a conservation benefit.

4.4. It will therefore be necessary to consider the above legislation when undertaking any river engineering works or those associated with the disruption of bank sides. This may include the installation of surface water drainage headwalls, bridges or landscape works, which should ideally avoid occupied habitat.

4.5. Opportunities for habitat enhancement do occur where over-shading by bank side vegetation has inhibited the development of any significant marginal vegetation that could provide cover and foraging habitat.

4.6. No evidence of otters was observed, however, given their potentially large range their occasional use of the site could not be ruled out. However, no places of rest or shelter were identified and it is therefore unlikely that this species represents a constraint.

4.7. Update survey for both are recommended where works are required within 5m (water voles) and 50m (otter) of banks sides.



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- Water Vole Dropping
- Rat Droppings
- L Water Vole Latrine
- F Water Vole Feeding Station
- P Mammal Prints

East Swindon Development Group

South Marston, Swindon

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WATER VOLE & OTTER SURVEY

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Figure 1

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